

TECHNICAL SPECIFICATION & RESEARCH BLUEPRINT

Cosmic-Ray Neutron Sensing (CRNS) for Dual Landslide & Flood Early Warning Systems

A Unified Physics-Informed Geotechnical, Hydrological, and Spatio-Temporal Machine Learning Framework for Real-Time Catchment & Hillslope Hazard Forecasting

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Executive Abstract: Slope failures (landslides) and catchment inundations (floods) have historically been treated as disconnected geohazard disciplines with isolated sensor networks. However, both hazards share the exact same physical driver: antecedent soil saturation within the root and sub-root vadose zone. Point sensors (TDR/capacitance) suffer from extreme spatial representation error and destroy delicate hillslope shear planes, while satellite radar (SMAP/Sentinel-1) cannot penetrate dense canopy or resolve rapid sub-daily dynamics. **Cosmic-Ray Neutron Sensing (CRNS)** bridges this critical scale gap by continuously measuring integrated soil moisture over a 150–240 m radius footprint down to 70 cm depth completely non-invasively. This document provides the exhaustive physics foundations, calibration mathematics, infinite slope geotechnical stability formulations, dynamic catchment flood models, spatio-temporal machine learning pipelines, and a phased end-to-end implementation blueprint for operational deployment.

1. Scientific Foundations of Cosmic-Ray Neutron Sensing

Cosmic-Ray Neutron Sensing (CRNS) is an advanced geophysical method developed to monitor terrestrial water storage at the mesoscale (field/hillslope scale). Primary galactic cosmic rays (predominantly high-energy protons) penetrate the Earth's magnetosphere and collide with atomic nuclei in the upper atmosphere (nitrogen and oxygen), generating a cascade of secondary cosmic-ray particles including fast neutrons. These fast neutrons propagate toward the Earth's surface, where they enter the soil matrix and interact with nuclei through elastic and inelastic collisions.

The Moderation Physics of Hydrogen: Among all terrestrial elements, hydrogen has an atomic mass nearly identical to that of a neutron (1.008 u vs 1.0086 u). Consequently, a neutron loses, on average, approximately 50% of its kinetic energy in a single collision with a hydrogen nucleus, whereas collisions with heavier elements (silicon, aluminum, oxygen) result in negligible momentum transfer. Hydrogen is therefore the dominant moderator of cosmic-ray fast neutrons. A fraction of these moderated (epithermal and fast, 0.2 eV to 100 keV) neutrons bounce back (albedo) into the atmosphere above the land surface. Because the density of hydrogen in the near-surface soil environment is governed almost entirely by liquid water, **the above-ground epithermal neutron flux is inversely proportional to soil water content (θ)**.

Footprint Geometry & Effective Depth

Unlike point-scale FDR/TDR probes that measure a few cubic centimeters, or satellite radiometry (e.g., SMAP) with a 9–36 km resolution, stationary CRNS sensors sample an intermediate footprint that matches hillslope geotechnical slip surfaces and sub-catchment drainage divides:

- **Horizontal Footprint (Radius R_{eff}):** Extensive Monte Carlo URANOS modeling (Köhli et al., 2015, 2021) demonstrates that 86% of detected neutrons originate within a radius of **130 m to 240 m** around the probe. The footprint radius expands slightly in dry, low-humidity air and contracts in saturated, humid conditions.
- **Vertical Sensing Depth (D_{eff}):** The effective measurement depth spans from **15 cm** in fully saturated soils up to **70 cm** in dry soils. This encompasses the root zone and immediate vadose zone where infiltration and perched water tables form.
- **Non-Invasive Integrity:** CRNS detectors are installed entirely above ground on a tripod or mast. They do not pierce, dig, or disturb the soil matrix. In active or metastable landslides, drilling access shafts for point sensors creates artificial preferential flow pathways that can trigger premature failure. CRNS avoids this risk entirely.

Mathematical Calibration Framework

Raw neutron counts recorded by stationary detectors (typically gas-filled ^3He or ^{10}B -lined proportional counters) must be systematically corrected for atmospheric pressure variations, ambient water vapor, solar intensity modulations, and biomass:

Correction Factor	Governing Formulation	Physical Significance
Pressure Correction (f_p)	$f_p = \exp((P - P_{\text{ref}}) / L_{\text{atm}})$	Compensates for variations in barometric shielding. $L_{\text{atm}} \approx 128\text{--}135 \text{ g/cm}^2$.
Water Vapor (f_v)	$f_v = 1 + 0.0054 \cdot (h_{\text{abs}} - h_{\text{ref}})$	Corrects for hydrogen atoms present in air column as humidity (h_{abs} in g/m^3).
Solar Intensity (f_i)	$f_i = I_{\text{ref}} / I_{\text{nmdb}}(t)$	Normalizes against galactic cosmic-ray flux variations using reference neutron monitors (e.g., Jungfraujoch).

Biomass Factor (f_bio)	$f_bio = 1 / (1 - 0.009 \cdot B_dry)$	Accounts for structural hydrogen in canopy vegetation and vegetation biomass growth.
Desilets N _■ Calibration	$\theta = [a_{\blacksquare} / ((N_corr/N_{\blacksquare}) - a_{\blacksquare}) - a_{\blacksquare}] \cdot (\rho_bulk / \rho_w) - w_lat - w_soc$	Universal neutron-to-moisture equation (Desilets et al., 2010). $a_{\blacksquare}=0.0808$, $a_{\blacksquare}=0.372$, $a_{\blacksquare}=0.115$.

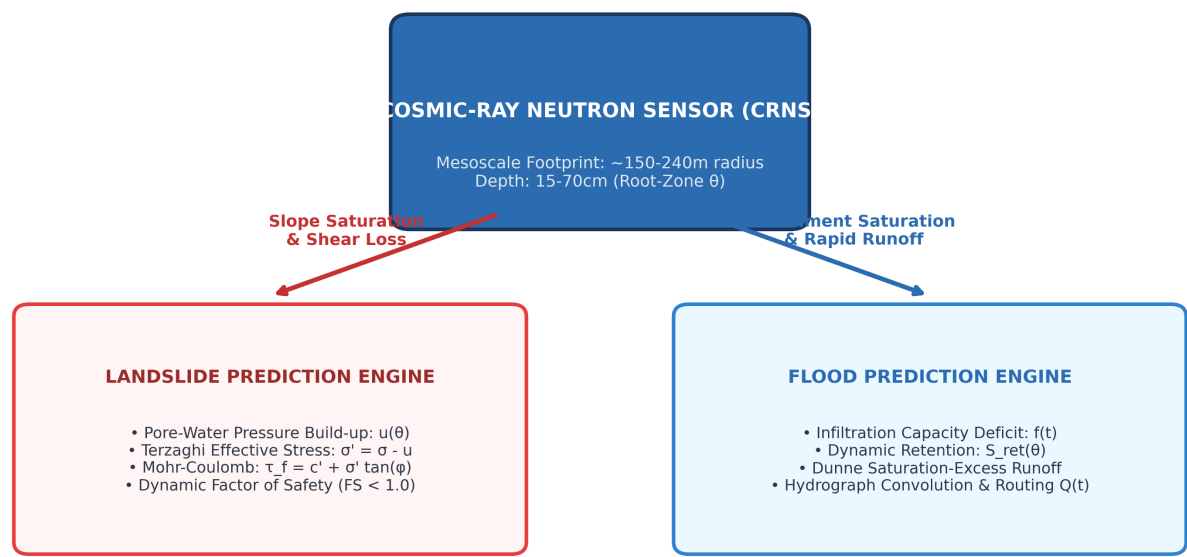


Figure 1: Conceptual physical coupling of Cosmic-Ray Neutron Sensing (CRNS) to Hillslope Geotechnical Stability (Landslides) and Catchment Hydrological Runoff (Floods).

2. Landslide Detection & Prediction Model Architecture

Rainfall-induced shallow landslides and translational debris flows are triggered primarily by the reduction of soil shear strength resulting from transient pore-water pressure generation during intense or prolonged precipitation. Conventional early warning systems rely exclusively on empirical Rainfall Intensity-Duration (I-D) thresholds. However, these empirical models produce unacceptable false-alarm rates because **a 50 mm storm falling on parched ground is completely absorbed without slope movement, whereas the same 50 mm storm falling on pre-saturated soil initiates catastrophic failure.**

Geomechanical Formulation: Dynamic Factor of Safety (FS)

We couple real-time CRNS volumetric water content (θ) directly into the geotechnical **Infinite Slope Stability Model**. The Factor of Safety (FS) is defined as the ratio of resisting shear strength (τ_f) to gravitational driving shear stress (τ_d):

$$FS = [c' + (\gamma_{bulk} \cdot z \cdot \cos^2\beta - u) \cdot \tan\phi'] / [\gamma_{bulk} \cdot z \cdot \sin\beta \cdot \cos\beta]$$

Where the variables are dynamically governed by the CRNS state variable θ :

- **c' (Effective Cohesion):** Geotechnical bond between soil grains (typically 3–15 kPa). In partially saturated states, apparent cohesion incorporates matric suction (van Genuchten SWCC), which rapidly drops to zero as $\theta \rightarrow \theta_{sat}$.
- **ϕ' (Internal Friction Angle):** Friction angle of the hillslope regolith (typically 28°–38°).
- **β (Slope Incline):** Hillslope topographical gradient extracted from high-resolution LiDAR DEM.
- **γ_{bulk} (Bulk Unit Weight):** Dynamically updated with CRNS water mass: $\gamma_{bulk} = \gamma_{dry} + (\theta \cdot \gamma_w)$. As water fills pores, driving mass increases.
- **u (Pore-Water Pressure):** Positive pore pressure develops when the soil column approaches critical saturation $S_r = \theta / \theta_{sat} > 0.85$: $u = m(\theta) \cdot \gamma_w \cdot z \cdot \cos^2\beta$, where $m(\theta)$ represents the normalized saturated water column depth.

Dynamic Threshold Adaptation (I-D Threshold Shifting)

Standard empirical thresholds define critical intensity as $I_{crit} = \alpha \cdot D^{(-\beta)}$. In our CRNS framework, the intercept α is formulated as a dynamic function of antecedent CRNS moisture content: $\alpha(\theta) = \alpha_{dry} \cdot [1 - (\theta / \theta_{sat})]^\gamma$. When the terrain is dry ($\theta = 0.15$), $\alpha(\theta)$ is high, requiring severe rainfall to trigger alerts. When CRNS measures saturation ($\theta > 0.40$), $\alpha(\theta)$ plummets, alerting authorities to potential slope failure even during low-to-moderate rain.

Hillslope State Variable	Typical Range	Source / Computation	Impact on Slope Stability
CRNS Volumetric Water Content (θ)	0.10 – 0.48 m ³ /m ³	Calibrated CRNS epithermal neutron flux	Governs effective stress and pore pressure buildup
Effective Cohesion (c')	4.0 – 12.0 kPa	Geotechnical borehole triaxial shear tests	Resisting force; vanishes under liquefaction
Internal Friction Angle (ϕ')	26° – 36°	In-situ direct shear tests	Frictional resistance along slip surface
Slope Gradient (β)	25° – 45°	LiDAR / UAV Digital Elevation Model (DEM)	Primary gravitational driving force
Regolith Depth (z)	1.2 – 2.5 m	Electrical Resistivity Tomography (ERT)	Defines failure plane overburden weight

Operational Landslide Alert Protocol

Alert Tier	Geotechnical FS	CRNS Saturation	Recommended Emergency Actions
GREEN (Normal)	$FS > 1.50$	$\theta / \theta_{sat} < 0.65$	Normal routine telemetry polling (1-hour intervals); baseline logging.
YELLOW (Advisory)	$1.25 < FS \leq 1.50$	$0.65 \leq \theta / \theta_{sat} < 0.80$	Increase polling to 15 mins; issue maintenance advisory for drainage culverts.
ORANGE (Warning)	$1.00 < FS \leq 1.25$	$0.80 \leq \theta / \theta_{sat} < 0.92$	Deploy road closures on vulnerable mountain passes; alert first responders.
RED (Imminent)	$FS \leq 1.00$	$\theta / \theta_{sat} \geq 0.92$	Trigger sirens, broadcast automated CAP emergency alerts, mandatory evacuation.

3. Flood Prediction Extension: Hydrological Coupling

Catchment flood forecasting models face their greatest source of error in quantifying **initial soil moisture conditions**. When hydrologists run continuous rainfall-runoff models, an over- or under-estimation of antecedent soil moisture leads to dramatic errors in hydrograph peak timing and flood stage height. Cosmic-Ray Neutron Sensing provides the missing physical link by supplying intermediate-scale catchment soil moisture that directly governs runoff generation mechanisms.

Runoff Physics: Horton vs. Dunne Infiltration Excess

Surface runoff is produced through two distinct physical regimes:

1. **Hortonian Runoff (Infiltration-Excess):** Occurs when rainfall intensity $i(t)$ exceeds the soil matrix infiltration capacity $f(t)$. While dominant in arid or impervious urban zones, it is rare in vegetated mountainous catchments.
2. **Dunne Runoff (Saturation-Excess):** The primary mechanism for flash floods in natural mountain valleys. Rain falls on hillslope soils whose storage capacity is already fully exhausted ($\theta = \theta_{\text{sat}}$). Because water cannot enter the saturated soil profile, 100% of additional precipitation immediately becomes overland flow and feeds the stream channel within minutes.

Dynamic SCS-CN Runoff Integration

In the USDA Soil Conservation Service (SCS-CN) framework, direct runoff depth Q_{excess} (mm) is calculated from precipitation P (mm) and maximum potential retention S_{ret} (mm):

$$Q_{\text{excess}} = (P - I_a)^2 / (P - I_a + S_{\text{ret}}) \quad \text{for } P > I_a$$

Where $I_a = \lambda \cdot S_{\text{ret}}$ is initial abstraction (interception + depression storage, $\lambda \approx 0.15\text{--}0.20$). In our implementation, rather than relying on static AMC (Antecedent Moisture Condition I, II, III) tables, **S_{ret} is dynamically updated every hour directly from the CRNS moisture deficit:**

$$S_{\text{ret}}(t) = S_{\text{max}} \cdot [1 - (\theta_{\text{CRNS}}(t) / \theta_{\text{sat}})]$$

When the catchment soil is dry ($\theta = 0.12$), S_{ret} is maximized (e.g., 110 mm), absorbing high rainfall volumes without runoff. When CRNS indicates antecedent saturation ($\theta = 0.44$), S_{ret} collapses toward 0 mm, converting almost all rainfall directly into surging surface runoff.

Streamflow Routing & Peak Discharge Convolution

The generated runoff depth series $Q_{\text{excess}}(t)$ is convoluted with the catchment synthetic unit hydrograph (UH) or routed through a 1D kinematic wave routing scheme to forecast downstream streamflow $Q(t)$ (m^3/s):

$$Q(t) = Q_{\text{base}} + \int Q_{\text{excess}}(\tau) \cdot u(t - \tau) d\tau$$

Where $u(t)$ is the impulse response function parameterized by the catchment area (A_{km^2}), time of concentration (T_c), and time to peak (T_p). This provides local civil protection authorities with a **6 to 48-hour forward-looking lead time** for downstream flood warnings.

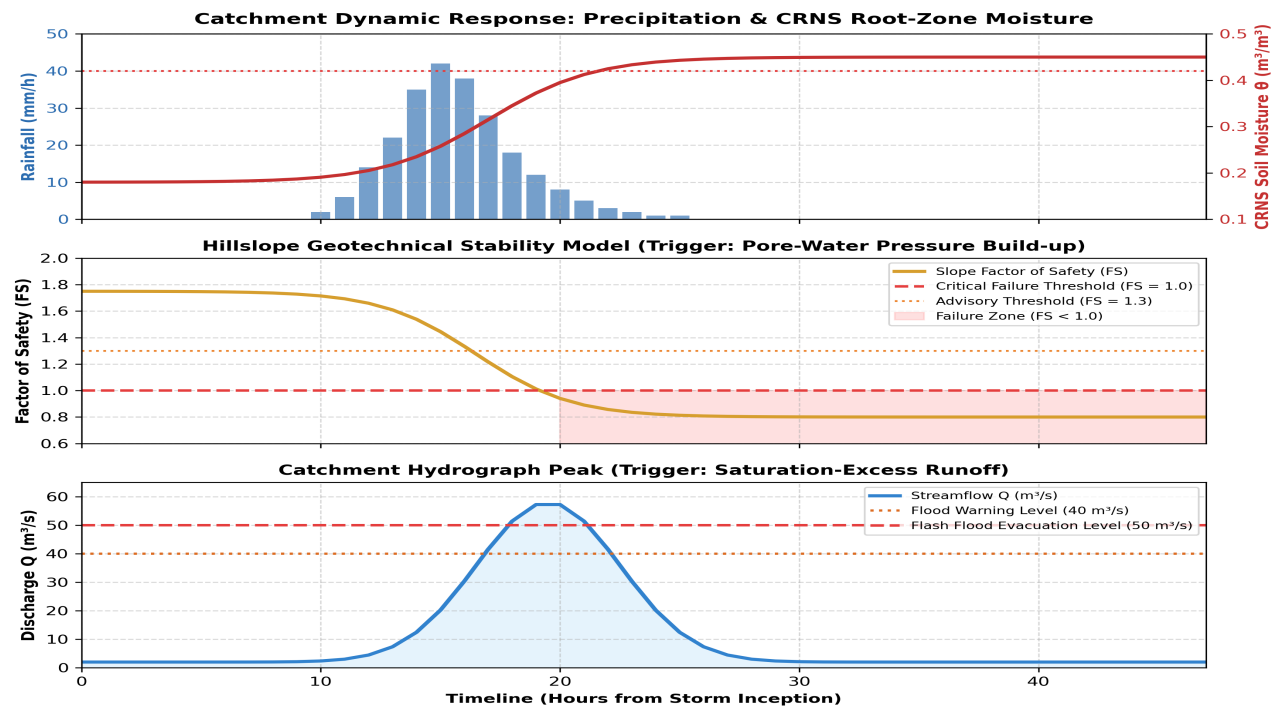


Figure 2: Dual-Hazard Simulation: 48-hour storm response showing simultaneous CRNS root-zone saturation, hillslope Factor of Safety (FS) collapse below 1.0, and downstream flood discharge hydrograph surge.

4. Spatio-Temporal Machine Learning Architecture

While physical geotechnical (FS) and hydrological (SCS-CN) models provide indispensable mechanistic ground-truth, they can be computationally demanding for regional real-time deployment and require rigorous soil property calibration. To maximize operational reliability, our system deploys a **Dual-Branch Spatio-Temporal Deep Learning Model (ConvLSTM / Multi-Task LSTM)** that consumes CRNS telemetry, meteorological feeds, and spatial terrain grids simultaneously.

Feature Engineering Matrix

The machine learning pipeline digests dynamic time-series coupled with static spatial terrain layers:

Category	Feature Name	Temporal / Spatial Resolution	Predictive Value / Purpose
CRNS Moisture	θ_{crns} , $d\theta/dt$ (1h, 6h, 24h)	Hourly / 150m footprint	Core antecedent saturation state and wetting front infiltration velocity
Hyetograph	Rain_cum (1h, 3h, 6h, 24h, 72h)	Hourly / Station or Radar (QPE)	Direct trigger of destabilizing pore pressure and runoff volume
Antecedent Index	API (Antecedent Precipitation Index)	Daily exponential decay	Quantifies cumulative hydrological memory across preceding 30 days
Topography (Static)	Slope, Aspect, Curvature, TWI	1m – 5m LiDAR DEM	Topographic Wetness Index (TWI) maps sub-surface convergence zones
Geology / Soil	Lithology, Grain size, Hydraulic K_sat	Vector GIS / Borehole database	Governs drainage capacity and regolith shear strength limits

Multi-Task Neural Network Architecture

A recurrent backbone (LSTM or GRU) extracts temporal sequence representations from the last 72 hours of hourly data. The shared latent embedding feeds two specialized prediction heads:

- **Head 1 (Landslide Classifier):** Dense layers with Sigmoid activation outputting probability $P(\text{Landslide Failure} \in [0, 1])$ over the next 12 hours. Trained using Focal Loss to combat extreme class imbalance (landslides are rare events).
- **Head 2 (Flood Discharge Regressor):** Linear head predicting downstream river flow $Q(t + \Delta t)$ across 6h, 12h, 24h, and 48h forward horizons. Trained using Nash-Sutcliffe Efficiency (NSE) Loss combined with Huber Loss.

5. Operational System Architecture & IoT Hardware

Deploying an early warning system in rugged, high-relief mountainous topography requires an autonomous, hardened, low-power sensor node design with robust failover telemetry.

Subsystem	Component / Hardware	Technical Specification	Operational Function
Neutron Detector	CRS-2000 / Hydroinnova or Quaesta Probe	Moderated + Bare ³ He or ¹⁰ B-lined proportional counters	Counts thermal (<0.5 eV) and epithermal (0.5 eV - 100 keV) neutrons
Meteorological Hub	All-in-one ultrasonic weather station	Barometric pressure (±0.1 hPa), Air temp, RH, Tipping bucket rain gauge	Required for real-time f_p and f_v corrections and rainfall tracking
Ground Reference	Piezometer & Tiltmeter array	Vibrating wire pore-pressure gauge + MEMS 3-axis inclinometer	Ground-truth validation for pore pressure and micro-movement
Downstream Gauge	Ultrasonic / Radar water level sensor	Range 0–15m, accuracy ±2mm, contactless	Real-time streamflow stage height and discharge verification
Power System	Solar MPPT + LiFePO4 battery bank	100W mono-crystalline panel + 12V 100Ah battery (14 days autonomy)	Continuous off-grid operation during extended storm cloudy periods
Telemetry Layer	CRSN / 4G LTE-M / Satellite Iridium	Cognitive Radio TV White Space / LoRaWAN mesh + Iridium SBD failover	Guaranteed packet delivery even when terrestrial cellular towers collapse

Telecommunications in Extreme Topography (CRSN Angle)

When standard cellular infrastructure fails during typhoons or landslides (downed poles, power outages), our design incorporates **Cognitive Radio Sensor Network (CRSN)** technology. In deep mountain gorges, standard 2.4 GHz or 915 MHz signals suffer severe diffraction losses from cliffs and wet foliage. CRSN nodes opportunistically sense and transmit across vacant **TV White Space (VHF/UHF, 470–698 MHz)**. These lower frequencies propagate over ridges and through dense tree cover, relaying high-priority emergency packets to a satellite uplink or microwave base station on the ridge.

6. Project Implementation Plan & Operational Roadmap

Deploying the CRNS Dual-Hazard Early Warning System is structured across four progressive phases spanning a 12-month timeline:

Phase & Timeline	Key Milestones & Deliverables	Validation Criteria
Phase 1: Months 1–3 Site Characterization & Sensor Calibration	• Topographic LiDAR survey & soil sampling (18 points in footprint). • Oven-dry gravimetric moisture, bulk density, lattice water, SOC analysis. • Installation of CRNS probe, meteorological station, and piezometers. • Derivation of site-specific N calibration constant via CRNPy.	Calibrated CRNS θ RMSE < 0.025 m ³ /m ³ against lab gravimetric samples.
Phase 2: Months 4–6 Telemetry & Pipeline Deployment	• Commissioning of primary LoRaWAN/cellular & CRSN/Satellite failover. • Automated ingestion microservice with quality control & spike filtering. • Connection to NMDB (Neutron Monitor Database) for real-time solar intensity f_i. • Deployment of downstream ultrasonic streamflow radar gauge.	99.8% telemetry uptime; latency from pulse to database < 90 seconds.
Phase 3: Months 7–9 Physics & ML Engine Tuning	• Parametrization of Infinite Slope Stability model (c', ϕ', β, z). • Dynamic SCS-CN catchment retention curve configuration. • Multi-task LSTM model training on historical regional disaster events. • Receiver Operating Characteristic (ROC) & False Positive Rate tuning.	Landslide ROC-AUC > 0.91; Flood Peak Discharge NSE > 0.85.
Phase 4: Months 10–12 Operational EWS & Commissioning	• Integration with municipal Civil Protection Common Alerting Protocol (CAP). • Multi-tier dashboard for emergency dispatchers. • Simulated full-scale disaster drill & siren activation. • Handover of system operations manual and maintenance protocols.	Automated alerts dispatched to emergency response teams in < 3 minutes.

Core Python Implementation Architecture

Below is the operational microservice blueprint orchestrating the ingestion, calibration, slope geotechnical safety factor calculation, and flood runoff generation in real time:

```
# Unified Early Warning Real-Time Pipeline
import numpy as np
from crns_engine import calibrate_neutron_counts
from geotechnical import infinite_slope_stability
from hydrology import dynamic_scs_cn_discharge

def process_telemetry_packet(packet):
    # 1. Atmospheric & Solar Corrections
    theta = calibrate_neutron_counts(
        raw_n=packet['counts'],
        pressure=packet['pressure'],
        temp=packet['temp'],
        rh=packet['rh'],
        n0=1240.0
    )
    # 2. Geotechnical Landslide Factor of Safety
    fs = infinite_slope_stability(
        theta=theta,
        slope_deg=34.5,
        depth=1.8,
        c_kpa=6.5,
        phi_deg=31.0
    )
    # 3. Dynamic Catchment Flood Runoff & Streamflow
    q_m3s = dynamic_scs_cn_discharge(
        rainfall_mm=packet['rain_1h'],
        theta=theta,
        catchment_area_km2=28.4,
        time_to_peak=3.5
    )
    # 4. Multi-Hazard Alerting Engine
    if fs < 1.0 or q_m3s > 50.0:
        dispatch_cap_alert(level='RED', fs=fs, discharge=q_m3s)
    elif fs < 1.25 or q_m3s > 35.0:
        dispatch_cap_alert(level='ORANGE', fs=fs, discharge=q_m3s)
    return {'theta': theta, 'fs': fs, 'q_m3s': q_m3s}
```

7. Summary & Recommendations

Cosmic-Ray Neutron Sensing delivers a transformative paradigm shift for disaster risk reduction in mountainous regions. By continuously tracking intermediate-scale vadose zone moisture without disturbing unstable slopes, it simultaneously solves the pore-water pressure trigger of landslides and the antecedent saturation deficit governing catastrophic flash flooding. Combined with edge telemetry, physics-informed stability equations, and spatio-temporal AI models, this system provides unprecedented reliability and actionable evacuation lead times to save lives and protect critical infrastructure.